UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 003 · Types of electrodes · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·5	Deck 003, slide 5 (rendered .pptx)
T003 00:07:50	AV-summary transcript 003, segment start
PDF p.57	Older consolidated Unit 2 slide PDF (8enterprise), page 57
ESA Dec-23 2d	Past paper: ESA Dec 2023, question 2.d
ESA Jan-25 s1 2b	Past paper: ESA Jan 2025 set 1, question 2.b

Contents

1. Part A · Formula sheet
2. 1 · Types of electrodes: the map
3. 2 · Metal–metal ion electrode
4. 3 · Metal–metal insoluble salt–anion electrode
5. 4 · Gas electrode
6. 5 · Amalgam electrode
7. 6 · Oxidation–reduction electrodes and the quinhydrone electrode
8. 7 · Ion-selective (membrane) electrode
9. Part B · Symbol glossary

A Formula sheet

1 Types of electrodes: the map

E1.1 · Nernst equation for any electrode (reduction form) ESA ★★

$$E = E^\circ - \frac{2.303RT}{nF}\log\frac{[\text{products}]}{[\text{reactants}]} = E^\circ - \frac{0.0591}{n}\log\frac{[\text{products}]}{[\text{reactants}]}\ \ (298\ \text{K})$$

GIVES The single-electrode (reduction) potential at any concentration.

APPLIES Write the electrode reaction as a reduction first; pure solids and pure metals count as 1, gases enter as partial pressure in atm.

UNITS V

NOTE $2.303RT/F = 0.0591$ V at 298 K (0.05914 with $R = 8.314$, $F = 96485$).

T003 00:01:38 · T003 00:07:50

2 Metal–metal ion electrode

E2.1 · Metal–metal ion electrode ESA ★

$$E_{\mathrm{M^{n+}/M}} = E^\circ_{\mathrm{M^{n+}/M}} + \frac{2.303RT}{nF}\log[\mathrm{M^{n+}}]$$

GIVES Potential of $\mathrm{M|M^{n+}}$.

APPLIES $\mathrm{M^{n+} + ne^- \rightleftharpoons M}$; pure metal, activity 1.

UNITS V

NOTE Same as $E^\circ - \frac{0.0591}{n}\log\frac{1}{[\mathrm{M^{n+}}]}$: the + sign comes from inverting the log.

S003·3 · T003 00:02:15

3 Metal–metal insoluble salt–anion electrode

E3.1 · Silver–silver chloride electrode ESA ★

$$E_{\mathrm{Ag/AgCl/Cl^-}} = E^\circ_{\mathrm{Ag/AgCl/Cl^-}} - \frac{0.0591}{1}\log[\mathrm{Cl^-}]$$

GIVES Potential of $\mathrm{Ag|AgCl(s)|Cl^-}$.

APPLIES $\mathrm{AgCl + e^- \rightleftharpoons Ag + Cl^-}$; $n = 1$.

UNITS V

NOTE Reversible to $\mathrm{Cl^-}$: E falls as $[\mathrm{Cl^-}]$ rises. $E^\circ \approx 0.222$ V (given data).

S003·4 · T003 00:04:57

E3.2 · Calomel electrode (same type) ESA ★

$$\mathrm{Hg_2Cl_2 + 2e^- \rightleftharpoons 2Hg + 2Cl^-}, \qquad E = E^\circ - \frac{0.0591}{2}\log[\mathrm{Cl^-}]^2 = E^\circ - 0.0591\log[\mathrm{Cl^-}]$$

GIVES Potential of $\mathrm{Hg|Hg_2Cl_2(s)|KCl}$.

APPLIES Reference electrode of class 004.

UNITS V

NOTE SCE (saturated KCl) 0.2422 V, 1 M 0.2824 V, 0.1 M 0.3358 V at 298 K (consolidated PDF).

NOT ON CURRENT SLIDES S003·4 · PDF p.33

4 Gas electrode

E4.1 · Hydrogen electrode ESA ★★

$$E_{\mathrm{Pt/H_2/H^+}} = E^\circ_{\mathrm{Pt/H_2/H^+}} - \frac{0.0591}{2}\log\frac{p_{\mathrm{H_2}}}{[\mathrm{H^+}]^2}$$

GIVES Potential of $\mathrm{Pt|H_2(p)|H^+}$.

APPLIES $\mathrm{2H^+ + 2e^- \rightleftharpoons H_2}$; $n = 2$; $E^\circ = 0$ by convention.

UNITS V

NOTE SHE: $p_{H_2} = 1$ atm, $[\mathrm{H^+}] = 1$ M, so the log term is 0 and $E = E^\circ = 0$.

S003·5 · T003 00:07:50 · T003 00:08:38

E4.2 · Hydrogen electrode in terms of pH ESA ★★

$$E = -\frac{0.0591}{2}\log p_{\mathrm{H_2}} - 0.0591\,\mathrm{pH} \qquad (p_{\mathrm{H_2}} = 1\text{ atm}:\ E = -0.0591\,\mathrm{pH})$$

GIVES The pH dependence of the hydrogen electrode.

APPLIES From E4.1 with $E^\circ = 0$ and $\mathrm{pH} = -\log[\mathrm{H^+}]$.

UNITS V

NOTE A hydrogen electrode can therefore measure pH.

ADDED FOR CLARITY S003·5

E4.3 · Chlorine electrode ESA ★

$$\mathrm{Cl_2 + 2e^- \rightleftharpoons 2Cl^-}, \qquad E = E^\circ - \frac{0.0591}{2}\log\frac{[\mathrm{Cl^-}]^2}{p_{\mathrm{Cl_2}}}$$

GIVES Potential of $\mathrm{Pt|Cl_2(p)|Cl^-}$.

APPLIES $n = 2$; $E^\circ = 1.36$ V (given data).

UNITS V

NOTE The slide names the chlorine electrode but gives only the hydrogen equation.

ADDED FOR CLARITY S003·5 · T003 00:06:45

5 Amalgam electrode

E5.1 · Lead amalgam electrode ESA ★

$$E_{\mathrm{Pb^{2+}/Pb\text{-}Hg}} = E^\circ_{\mathrm{Pb^{2+}/Pb\text{-}Hg}} - \frac{0.0591}{2}\log\frac{[\mathrm{Pb\text{-}Hg}]}{[\mathrm{Pb^{2+}}]}$$

GIVES Potential of $\mathrm{Pb-Hg|Pb^{2+}}$.

APPLIES $\mathrm{Pb^{2+} + 2e^- \rightleftharpoons Pb(Hg)}$; $n = 2$.

UNITS V

NOTE Unlike a pure metal, the amalgam's activity is not 1, so it stays in the log term.

S003·6 · T003 00:10:54 · T003 00:11:06

6 Oxidation–reduction electrodes and the quinhydrone electrode

E6.1 · Stannic–stannous redox electrode ESA ★★

$$E_{\mathrm{Pt/Sn^{4+}/Sn^{2+}}} = E^\circ_{\mathrm{Pt/Sn^{4+}/Sn^{2+}}} - \frac{0.0591}{2}\log\frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$$

GIVES Potential of $\mathrm{Pt|Sn^{4+}, Sn^{2+}}$.

APPLIES $\mathrm{Sn^{4+} + 2e^- \rightleftharpoons Sn^{2+}}$; $n = 2$.

UNITS V

NOTE Only the ratio [reduced]/[oxidised] matters. $E^\circ = 0.15$ V (given data).

S003·7 · T003 00:13:31

E6.2 · Ferric–ferrous redox electrode

ESA ★★

$$\mathrm{Fe^{3+} + e^- \rightleftharpoons Fe^{2+}}, \qquad E = E^\circ - 0.0591 \log \frac{[\mathrm{Fe^{2+}}]}{[\mathrm{Fe^{3+}}]}$$

GIVES Potential of $\mathrm{Pt|Fe^{3+}, Fe^{2+}}$.

APPLIES $n = 1$; $E^\circ = 0.77$ V (given data).

UNITS V

NOTE The Pt indicator electrode of the potentiometric titration of ferrous iron (lab).

ADDED FOR CLARITY S003·7 · T003 00:13:05 · PDF p.67

E6.3 · Quinhydrone electrode

ESA ★★

$$E_{\mathrm{Pt/Q/QH_2}} = E^\circ_{\mathrm{Pt/Q/QH_2}} - \frac{0.0591}{2} \log \frac{[\mathrm{QH_2}]}{[\mathrm{Q}][\mathrm{H^+}]^2}, \qquad E^\circ_{\mathrm{Pt/Q/QH_2}} = +0.699 \text{ V}$$

GIVES Potential of $\mathrm{Pt|Q, QH_2}$.

APPLIES $\mathrm{Q + 2H^+ + 2e^- \rightleftharpoons QH_2}$; $n = 2$.

UNITS V

NOTE $\mathrm{H^+}$ is a reactant, so it sits in the denominator, squared.

S003·8 · T003 00:15:20

E6.4 · Quinhydrone electrode in terms of pH

ESA ★★

$$[\mathrm{Q}] = [\mathrm{QH_2}] \implies E = 0.699 + 0.0591 \log[\mathrm{H^+}] = 0.699 - 0.0591\,\mathrm{pH}$$

GIVES The pH dependence used to measure pH.

APPLIES Quinhydrone dissolved (1 : 1); pH below about 8.

UNITS V

NOTE $\frac{0.0591}{2} \log[\mathrm{H^+}]^2 = 0.0591 \log[\mathrm{H^+}]$.

ADDED FOR CLARITY S003·8

7 Ion-selective (membrane) electrode

E7.1 · Glass electrode (pH-sensitive ISE)

ESA ★★

$$E_G = E^\circ_G + 0.0591 \log_{10}[\mathrm{H^+}] = E^\circ_G - 0.0591\,\mathrm{pH}$$

GIVES Potential of the glass electrode.

APPLIES Glass membrane selective to $\mathrm{H^+}$; 298 K.

UNITS V

NOTE E°_G differs for each electrode and is found by calibration with a buffer (Unit 3).

S003·9 · T003 00:17:47 · PDF p.56

E7.2 · Glass + calomel cell (PESU form) ESA ★★

$$E_{cell} = E_G - E_{cal} = E_G^{\circ} - 0.0591\,\mathrm{pH} - E_{cal}$$

GIVES	The emf when a glass electrode is coupled with a calomel electrode.
APPLIES	Calomel written on the left (anode), as in the consolidated PDF.
UNITS	V
NOTE	The pH numericals that use this are Unit 3 (class 36).

NOT ON CURRENT SLIDES PDF p.57 · PDF p.58 · ESA Jan-25 s1 2b

B Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
E	Electrode potential	Reduction potential of the electrode at the given concentrations	V	E1.1–E7.1
E°	Standard electrode potential	E with all activities 1 (1 M, 1 atm)	V	E1.1–E6.3
E_G, E°_G	Glass electrode potential	E°_G is a constant of each glass electrode	V	E7.1, E7.2
E_{cal}, E_{SCE}	Calomel potential	Saturated calomel 0.2422 V (PDF; papers use 0.2412–0.244 V)	V	E3.2, E7.2
n	Electrons transferred	From the balanced reduction reaction	—	E1.1–E6.3
R, T, F	Gas constant, temperature, Faraday	8.314 J/(K mol), 298 K, 96485 C/mol; $2.303RT/F = 0.0591$ V		E1.1, E2.1
[ ]	Concentration	Molar concentration (strictly activity)	mol/L	all
$p_{\mathrm{H_2}}$, $p_{\mathrm{Cl_2}}$	Partial pressure	Activity of a gas	atm	E4.1–E4.3
pH	pH	$-\log[\mathrm{H^+}]$	—	E4.2, E6.4, E7.1
Q, $\mathrm{QH_2}$	Quinone, hydroquinone	Oxidised and reduced forms in the quinhydrone electrode (not a reaction quotient)		E6.3, E6.4
Pb-Hg	Lead amalgam	Lead dissolved in mercury; activity below 1		E5.1
| and ||	Phase boundary, salt bridge	Single bar separates phases; double bar a salt bridge or junction		E7.2, W7.1

Remark. E° is the standard reduction potential of the electrode named in its subscript. Square brackets are molar concentrations (strictly activities); for a gas the partial pressure p (atm) takes their place. Q and $\mathrm{QH_2}$ are quinone and hydroquinone, not the reaction quotient.